GNN4EEG DGNN/RGNN/SparseDGNN & HetEmotionNet Reproduction Report

Table of Contents

1. Introduction

GNN4EEG is an open benchmark and software toolkit for graph-neural-network-based electroencephalography classification. It addresses a central limitation of conventional EEG classifiers: flattening multi-channel signals can weaken or remove spatial relations among scalp electrodes. GNN4EEG instead represents each EEG channel as a graph node and models channel relations as graph edges [1].

The FACED benchmark contains EEG recordings from 123 participants watching 28 emotion-elicitation videos. The paper defines cross-9, cross-2, intra-9, and intra-2 tasks, four graph-based models, and three validation protocols. Cross-9 is the most difficult setting because the model must classify nine emotions for participants who are completely absent from training.

This revised report integrates verified outputs from both completed notebooks: (1) the DE/LDS exact-NCV notebook for DGCNN, RGNN, and SparseDGCNN, and (2) the subject-wise two-stream HetEmotionNet-style notebook. It follows the organizational structure and evidence discipline of the supplied Group 09 reproduction-report template [7].

2. Why GNN4EEG Was Selected

Selection criterion	Justification
Dataset scale	FACED contains 123 participants, providing a larger cross-subject benchmark than many affective-EEG datasets.
Graph suitability	EEG electrodes naturally form graph nodes, while anatomical, statistical, or learned relations form edges.
Task relevance	Cross-9 measures subject-independent recognition across nine emotions and exposes subject specificity.
Model breadth	The benchmark compares adaptive, signed, sparse, and heterogeneous graph-learning strategies.
Evaluation value	CV, FCV, and NCV show how epoch and hyperparameter selection affect reported generalization.
Reproduction value	Both completed notebooks contain fold construction, leakage checks, hyperparameter selection, checkpoint restoration, and final test outputs.

3. GNN4EEG Framework and Model Architecture

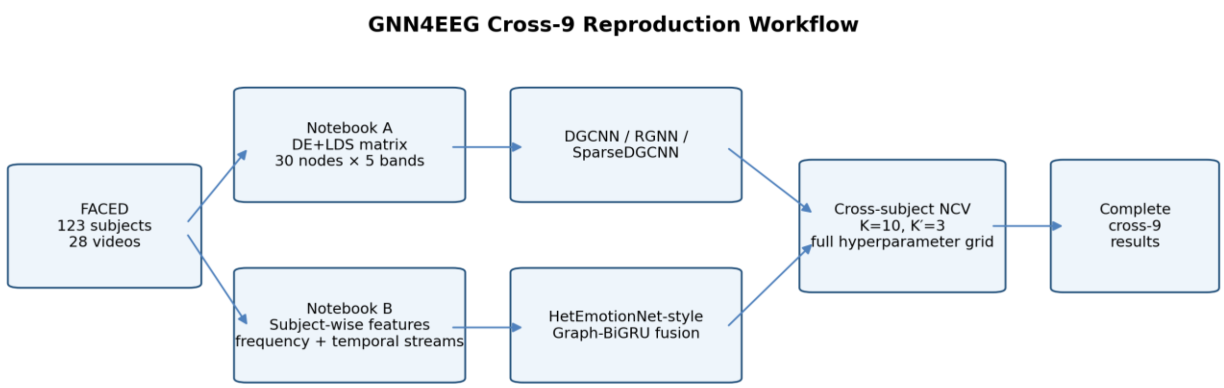

Fig. 1. Workflow of the two completed GNN4EEG cross-9 notebooks.

3.1 Benchmark construction and preprocessing

The paper retains the final 30 seconds of each video, removes the final two channels, and represents every one-second window by five differential-entropy frequency features smoothed with linear dynamic systems. This produces 30 graph nodes and five node features per sample. The completed main notebook confirms the canonical matrix (123 × 840 × 150) and graph input (103,320 × 30 × 5).

Benchmark element	Specification
Participants	123
Videos per participant	28
Retained duration	30 seconds per video
Graph nodes	30 EEG channels
Node features	Delta, theta, alpha, beta, and gamma
Classes	Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness
One-second graph samples	$123 \times 28 \times 30 = 103{,}320$

3.2 Graph representation

For an EEG graph G = (V, E), V is the set of channels and E contains channel relations. With node-feature matrix X, adjacency A, and degree matrix D, one graph-convolution layer can be expressed as H(l+1) = σ(D^(-1/2) A D^(-1/2) H(l) W(l)). The four models differ mainly in how A is initialized, learned, constrained, or combined with recurrent processing [1].

3.3 DGCNN

DGCNN learns a complete symmetric adjacency matrix during training. The notebook applies two graph-propagation layers, projects negative learned edges to zero after optimization, flattens the node embeddings, and classifies the resulting graph representation.

3.4 RGNN

RGNN uses a signed distance-based graph and activates NodeDAT domain-adversarial training for cross-subject evaluation. Gradient reversal encourages subject-invariant representations. In the completed run, RGNN consistently selected the largest hidden dimension and required substantially more epochs than the other models.

3.5 SparseDGCNN

SparseDGCNN retains the DGCNN-style classifier but applies proximal soft-thresholding to the learned adjacency. This encourages a sparse channel graph and implements the intended forward-backward optimization principle more directly than a simple L1 penalty.

3.6 HetEmotionNet-style implementation

The paper describes HetEmotionNet as a two-stream heterogeneous graph recurrent network. The completed notebook implements an EEG-only adaptation: a 30×5 frequency stream and a 30×30 temporal stream undergo graph propagation and bidirectional GRU processing, followed by projection, fusion, and nine-class classification. Because the architecture and feature source differ from the official implementation, the result is labelled HetEmotionNet-style rather than an exact HetEmotionNet reproduction.

3.7 Validation protocols

Protocol	Selection and reporting rule	Reliability implication
CV	Each fold reports its best validation epoch; results are averaged.	Can be optimistic because folds may select different epochs.
FCV	One epoch is selected from the average validation curve and used for all folds.	Reduces fold-specific selection but still reports validation performance.
NCV	Inner folds select hyperparameters and epoch; an untouched outer fold is tested once.	Most defensible estimate of subject-independent generalization.

4. Original Paper Results on FACED

The GNN4EEG paper reports the following cross-9 accuracies. NCV is the principal comparison because both completed notebooks use nested cross-validation.

Model	CV (%)	FCV (%)	NCV (%)
DGCNN	30.8	28.9	26.7
RGNN	34.0	32.1	31.3
SparseDGCNN	30.7	29.0	27.7
HetEmotionNet	32.5	30.8	30.8

RGNN is the strongest published cross-9 model under all three protocols. The paper attributes its cross-subject advantage to NodeDAT and reports that SparseDGCNN behaves similarly to DGCNN [1].

5. Reproduction Scope

Component	Reproduction target
Dataset	FACED processed features
Task	Cross-subject nine-class emotion classification
Samples	One one-second graph sample for each retained second
Nodes and features	30 channels and five frequency-band features
Notebook A	DGCNN, RGNN, and SparseDGCNN on DE/LDS input
Notebook B	EEG-only HetEmotionNet-style two-stream graph recurrent model
Validation	10 outer folds and 3 inner folds
Hyperparameter grid	Hidden dimensions 20/40/80; learning rates 1e-4/1e-3/1e-2
Primary metric	Outer-test accuracy; macro-F1 is the common balance-sensitive metric
Evidence boundary	Only numerical outputs actually stored in the completed notebooks are reported

6. Exact Reproduction Requirements and Availability

Updated evidence status

Verified ten-fold outputs are now available for all four evaluated models. The remaining limitation is not missing results; it is model and pipeline identity relative to the official GNN4EEG implementation.

Requirement	Availability	Reproduction decision
GNN4EEG paper and benchmark table	Available	Used as the published reference.
FACED DE/LDS matrix	Verified in Notebook A	Loaded from FACED_dataset_9_labels.mat, key de_lds.
Completed DGCNN/RGNN/Sparse outputs	Available	All 30 outer-model results were included.
Completed HetEmotionNet-style outputs	Available	All ten folds, pooled predictions, and class metrics were included.
Official subject identities	Unavailable	Notebook A uses contiguous subject blocks; Notebook B uses deterministic seed-42 splits.
Original random seed and trained weights	Unavailable	Models were restored or trained from notebook checkpoints; identity matching is impossible.
Official model code identity	Not established	Notebook implementations are treated as paper-aligned reimplementations.
Official HetEmotionNet feature/graph pipeline	Not reproduced exactly	PSD-path features, clip-level adjacency, and Graph-BiGRU fusion were used.
Hardware match	Unavailable	Both completed notebooks used a Tesla T4; the paper reports an RTX 3090.

7. Notebook Design

7.1 Completed DGCNN/RGNN/SparseDGCNN notebook

The completed resumed-cross9 notebook follows this sequence: package and dataset validation → MAT loading → canonical graph conversion → class and shape checks → subject-wise z-score normalization → deterministic subject-disjoint outer and inner folds → model definitions → full inner validation curves for epochs 1–100 → weighted inner-accuracy selection → fresh outer retraining → test metrics and aggregate confusion matrices → checkpoint and result export.

The notebook disables early stopping, stores every candidate epoch, validates restored checkpoint metadata, and resumes completed folds without repeating finished work. Its outputs contain 30 completed outer-fold records: ten each for DGCNN, RGNN, and SparseDGCNN.

7.2 Completed HetEmotionNet-style notebook

The completed resume notebook restores subject-wise feature files, canonicalizes 123 subjects, applies subject-wise normalization, builds frequency and temporal streams, constructs 3,444 clip-level mutual-information graphs, executes 10×3 nested subject folds, resumes the hyperparameter grid, trains ten final outer models, and exports fold metrics, pooled predictions, confusion matrix, classification report, and checkpoint archives.

The notebook detected 274 completed runs and nine active checkpoints at resume start. Final inspection confirmed ten completed final-model checkpoints and no incomplete final models.

8. Experimental Configuration

8.1 Shared cross-9 configuration

Setting	Value
Participants / classes	123 / 9
Total graph samples	103,320
Outer / inner folds	10 / 3
Outer test-subject pattern	12 subjects in folds 1–9; 15 subjects in fold 10
Hidden-dimension grid	20, 40, 80
Learning-rate grid	0.0001, 0.001, 0.01
Maximum candidate epochs	100
Batch size	256
L1 / L2 for cross-9	0.005 / 0.005
Subject normalization	Per-subject z-score
Random seed	42
Execution device	Tesla T4 GPU

8.2 Main notebook configuration

Setting	Value used	Evidence/status
Data source	FACED_dataset_9_labels.mat; de_lds	Verified notebook output
Canonical data	123 × 840 × 150	Verified
Model input	103,320 × 30 × 5	Verified
Models	DGCNN, RGNN, SparseDGCNN	All completed
GNN layers	2	Notebook configuration
Dropout	0.5	RGNN/classifier configuration
NodeDAT	Enabled for RGNN	Notebook configuration
Early stopping	Disabled	Required for complete epoch selection

Setting	Value used	Evidence/status
Mixed precision	Enabled	Notebook configuration
Software	Python 3.12.13; PyTorch 2.10.0+cu128	Notebook output

8.3 HetEmotionNet-style configuration

Setting	Value used	Evidence/status
Accepted subjects	123	Notebook output
Canonical feature array	123 × 28 × 30 × 30 × 5	After dropping two channels
Frequency stream	30 × 5 per second	Current second across five bands
Temporal stream	30 × 30 per clip	Thirty seconds averaged across bands
Graph cache	3,444 matrices of 30 × 30	One per subject-video clip
Adjacency	Mutual information; 10 bins; threshold 0.70	Notebook configuration
Global graph density	4.47% non-zero	Notebook output
Model	Two Graph-BiGRU streams + projection + fusion	Standalone adaptation
Optimizer	Adam	Notebook code
Software	Python 3.12.13; PyTorch 2.10.0+cu128	Notebook output

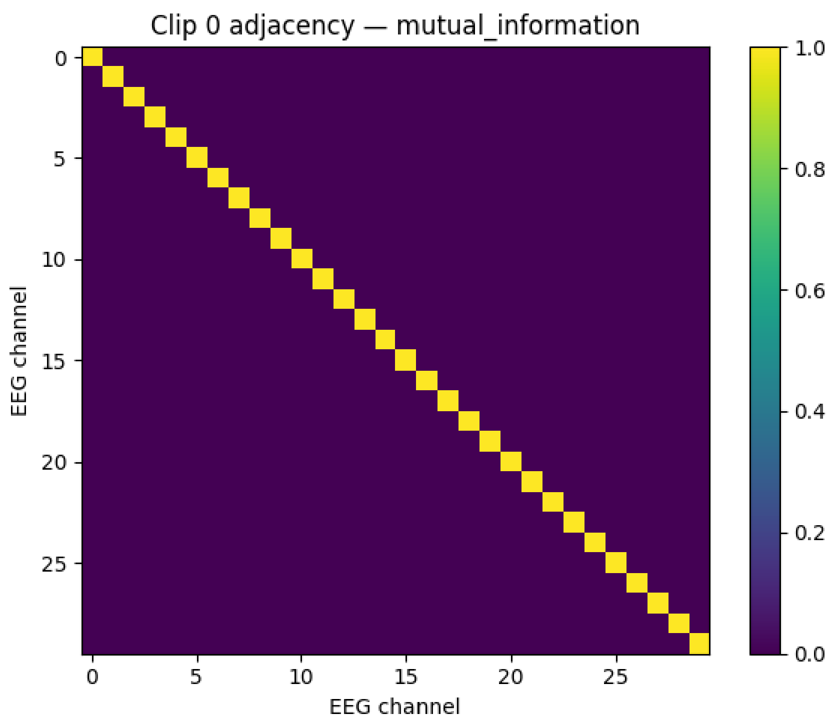

Fig. 2. Example clip-level mutual-information adjacency; this clip retains only self-loops at the 0.70 threshold.

8.4 Comparability limitations

1. The paper benchmark specifies DE features smoothed by LDS, while the HetEmotionNet-style manifest points to PSD feature files.
2. The official HetEmotionNet uses heterogeneous graph transformation; the notebook uses normalized graph propagation and bidirectional GRUs.
3. The HetEmotionNet-style notebook computes one graph per 30-second clip and reuses it for all 30 one-second samples.
4. Exact original subject identities, seed, trained weights, and model-version details are unavailable.
5. The main notebook is substantially more paper-aligned, but its model code is still a reconstruction rather than a verified byte-identical release build.

9. Evaluation Metrics

Metric	Meaning	Use in this report
Accuracy	Correct predictions divided by all predictions.	Primary comparison with the published NCV table.
Macro precision	Mean class precision over nine emotions.	Treats every class equally.
Macro recall	Mean class recall over nine emotions.	Shows average class sensitivity.
Macro-F1	Mean of nine class-specific F1 scores.	Common balance-sensitive metric across both notebooks.
Weighted F1	F1 weighted by class support.	Supplementary metric from Notebook A.
Per-class metrics	Precision, recall, and F1 for each emotion.	Available as pooled output for the HetEmotionNet-style notebook.
Confusion matrix	Row-normalized true versus predicted outcomes.	Shows error direction and class collapse.
Mean ± standard deviation	Average and variability across ten outer folds.	Measures cross-subject stability.

10. Reproduced Results

10.1 DGCNN, RGNN, and SparseDGCNN fold results

Fold	DGCNN Acc.	DGCNN F1	RGNN Acc.	RGNN F1	Sparse Acc.	Sparse F1
1	44.52	43.77	28.68	27.58	40.16	39.58
2	44.28	43.57	31.37	27.50	43.22	42.82
3	43.44	42.48	26.92	22.65	41.17	39.32
4	42.74	41.97	35.37	32.16	44.69	44.15
5	41.98	41.60	28.68	24.98	41.20	40.59
6	39.25	38.31	30.95	27.50	35.53	34.33
7	37.05	35.75	28.77	24.25	34.87	33.89
8	34.32	33.61	32.79	28.34	38.53	37.71
9	36.13	34.62	30.87	26.76	36.08	34.67
10	40.63	39.97	26.69	23.41	38.16	37.56

DGCNN achieved its highest fold accuracy in fold 1 (44.52%) and its lowest in fold 8 (34.32%). SparseDGCNN peaked in fold 4 (44.69%) and fell to 34.87% in fold 7. RGNN was consistently weaker, ranging from 26.69% to 35.37%. These ranges confirm that unseen-subject composition materially changes cross-9 difficulty.

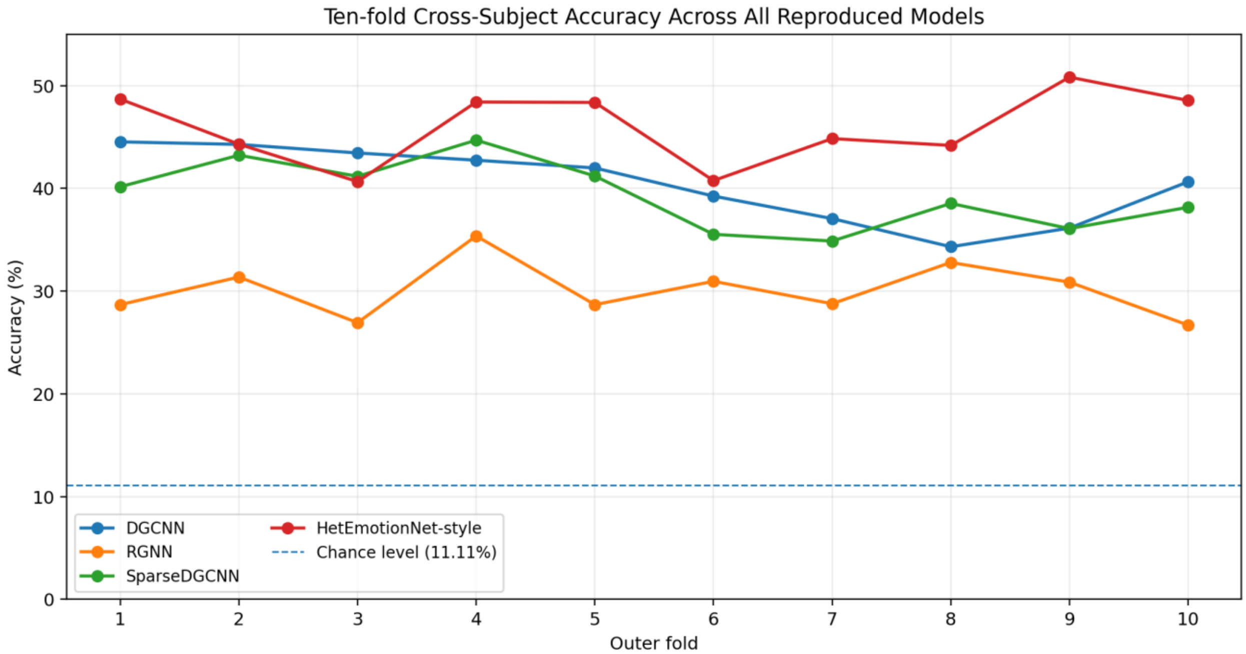

Fig. 3. Outer-fold accuracy for all four reproduced models. The dashed line marks nine-class chance performance.

10.2 Main-notebook aggregate results

Model	Accuracy, mean ± SD	Macro-F1, mean ± SD	Minimum–maximum accuracy	Pooled accuracy	Mean selected epoch
DGCNN	40.43 ± 3.61%	39.57 ± 3.78%	34.32–44.52%	40.44%	12.3
RGNN	30.11 ± 2.70%	26.51 ± 2.80%	26.69–35.37%	30.03%	75.7
SparseDGCNN	39.36 ± 3.31%	38.46 ± 3.52%	34.87–44.69%	39.33%	15.2

DGCNN was the strongest model in Notebook A, but SparseDGCNN was only 1.07 percentage points lower. Their confusion patterns are also similar: both recognize Anger and Neutral most reliably and struggle with Joy, Inspiration, and Sadness. The sparse constraint therefore changed the graph structure without producing a decisive accuracy advantage.

RGNN displays a different failure pattern. Its accuracy exceeds its macro-F1 by 3.60 percentage points, showing uneven class performance. The aggregate matrix concentrates predictions on Neutral, Anger, and Tenderness, while recall falls to 0.10 for Amusement, 0.14 for Inspiration, 0.16 for Sadness, and 0.05 for Joy. The model selected hidden dimension 80 and learning rate 0.001 in every fold, with a mean of 75.7 epochs, but the longer optimization did not translate into stronger generalization.

10.3 HetEmotionNet-style fold results

Outer fold	Hidden	Learning rate	Selected epoch	Accuracy (%)	Macro-F1 (%)
1	40	0.0100	3	48.68	48.55
2	20	0.0100	8	44.30	43.53
3	40	0.0100	9	40.65	40.75
4	40	0.0100	45	48.40	48.30
5	80	0.0100	20	48.36	47.71
6	40	0.0100	7	40.75	40.87
7	40	0.0100	13	44.84	44.88
8	80	0.0100	3	44.18	44.18
9	40	0.0001	21	50.82	50.57
10	40	0.0100	5	48.56	48.11

The HetEmotionNet-style model achieved the highest fold accuracy in fold 9 (50.82%) and the lowest in fold 3 (40.65%), a 10.17-point range. Hidden dimension 40 was selected in seven folds, and learning rate 0.01 was selected in nine folds. This preference differs from the paper's reported HetEmotionNet sensitivity at high learning rate, reinforcing that the notebook architecture is materially different.

10.4 HetEmotionNet aggregate and per-class results

Metric	Mean ± standard deviation	Minimum	Maximum
Accuracy	45.95 ± 3.53%	40.65%	50.82%
Macro precision	46.90 ± 3.42%	41.22%	50.82%
Macro recall	46.21 ± 3.46%	40.78%	50.88%
Macro-F1	45.74 ± 3.40%	40.75%	50.57%

Because fold 10 contains 15 subjects while the other folds contain 12, the pooled sample-level accuracy is 46.02%, slightly above the unweighted mean fold accuracy of 45.95%.

Emotion	Precision	Recall	F1	Support
Anger	0.42	0.43	0.43	11,070
Disgust	0.44	0.43	0.44	11,070
Fear	0.50	0.50	0.50	11,070
Sadness	0.44	0.36	0.39	11,070
Neutral	0.43	0.39	0.41	14,760
Amusement	0.53	0.67	0.60	11,070
Inspiration	0.43	0.42	0.43	11,070
Joy	0.55	0.51	0.53	11,070
Tenderness	0.40	0.46	0.42	11,070

Amusement is the strongest class (F1 = 0.60; recall = 0.67), followed by Joy (F1 = 0.53) and Fear (F1 = 0.50). Sadness is the weakest class (F1 = 0.39), while Neutral and Tenderness remain difficult despite adequate sample support.

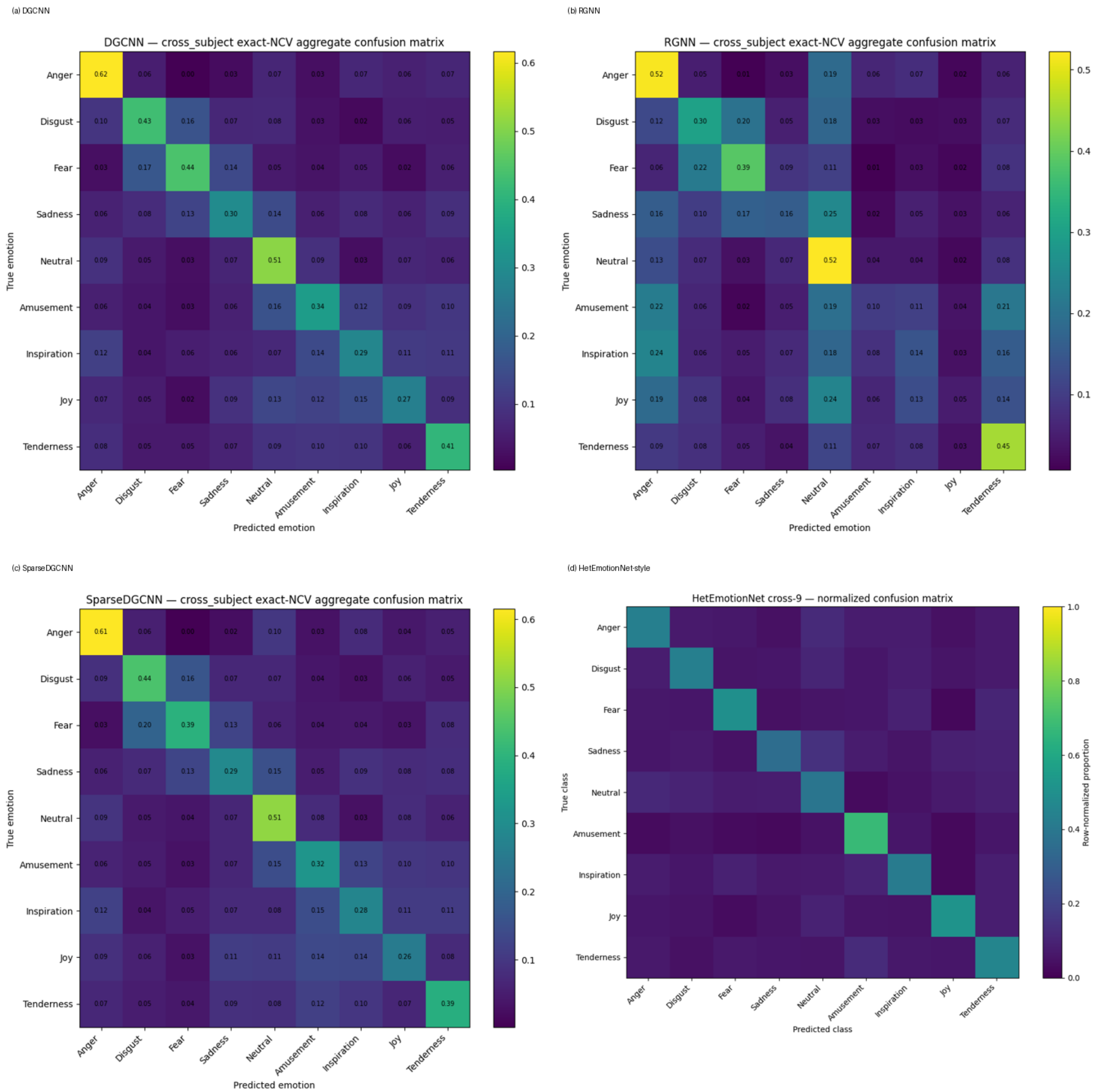

Fig. 4. Row-normalized aggregate confusion matrices from both completed notebooks.

10.5 Summary of the Complete Results

Rank	Model	Accuracy, mean ± SD	Macro-F1, mean ± SD	Pooled accuracy	Key result
1	HetEmotionNet-style	45.95 ± 3.53%	45.74 ± 3.40%	46.02%	Best overall and most balanced result.
2	DGCNN	40.43 ± 3.61%	39.57 ± 3.78%	40.44%	Best paper-aligned graph model.
3	SparseDGCNN	39.36 ± 3.31%	38.46 ± 3.52%	39.33%	Close to DGCNN; sparsity gave no clear gain.
4	RGNN	30.11 ± 2.70%	26.51 ± 2.80%	30.03%	Weakest and most class-imbalanced result.

Whole-results interpretation

All four models exceed the 11.11% chance level, so each learned non-trivial emotion-related EEG structure. The HetEmotionNet-style model is the strongest and most class-balanced, but it is also the least directly comparable with the official paper because its features, graph frequency, and recurrent architecture differ. Within the more paper-aligned notebook, DGCNN and SparseDGCNN form a close performance group, while RGNN fails to reproduce the published cross-subject advantage. The defensible conclusion is therefore successful reconstruction of the complete cross-9 evaluation pipeline, not verified superiority over the official implementations.

11. Original Versus Reproduced Results

Model	Published NCV	Reproduced NCV	Difference	Comparison status
DGCNN	26.7%	40.43 ± 3.61%	+13.73 pp	Paper-aligned reconstruction; code identity unverified
RGNN	31.3%	30.11 ± 2.70%	-1.19 pp	Closest numerical reproduction
SparseDGCNN	27.7%	39.36 ± 3.31%	+11.66 pp	Paper-aligned reconstruction; code identity unverified
HetEmotionNet-style	30.8%	45.95 ± 3.53%	+15.15 pp	Not directly comparable; architecture/features differ

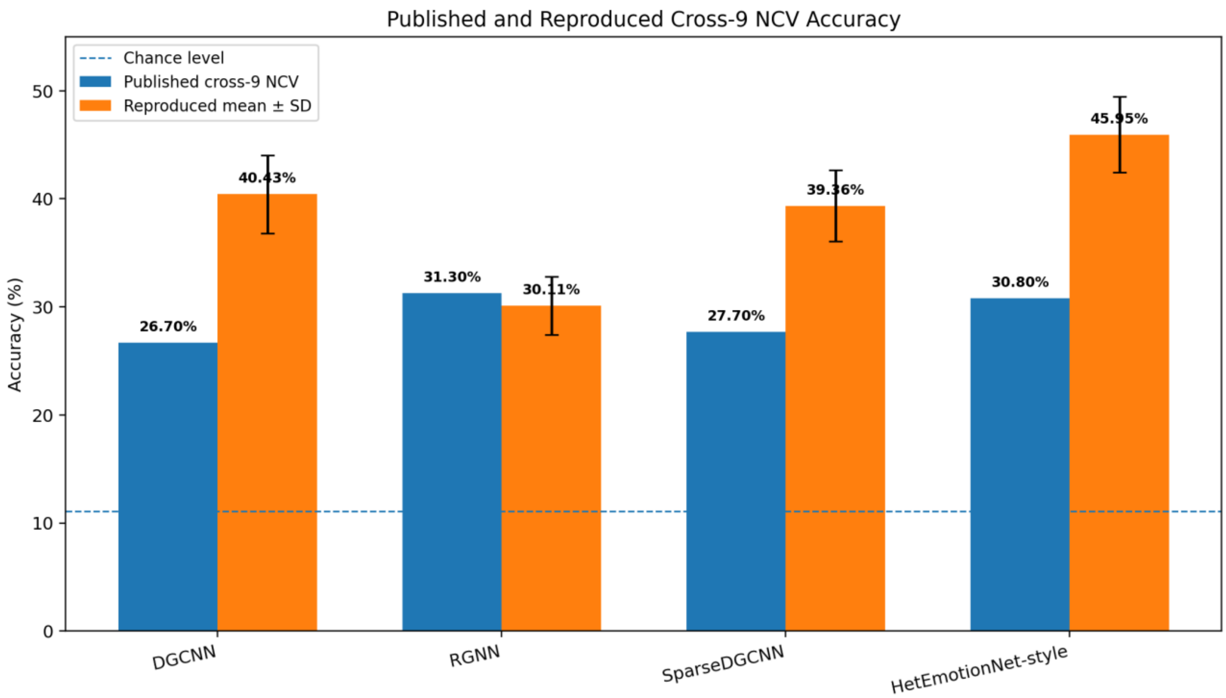

Fig. 5. Published and reproduced cross-9 NCV accuracy. Error bars show reproduced fold standard deviation.

RGNN is the only result close to the published value, finishing 1.19 percentage points lower. DGCNN, SparseDGCNN, and the HetEmotionNet-style model exceed the published table by 11.66–15.15 points. Differences of this magnitude cannot be treated as ordinary random variation; they indicate changes in model implementation, input processing, fold identity, or optimization. These values therefore describe the uploaded notebook implementations rather than verified improvements to the official GNN4EEG models.

12. Source and Implementation Differences

Difference	Published / official description	Notebook implementation	Possible effect
Fold identities	Exact subject IDs are not listed.	Notebook A uses contiguous 12-subject blocks; Notebook B uses deterministic seed-42 folds.	Changes unseen-subject difficulty.
Model code identity	Official toolkit implementations.	Paper-aligned independent notebook implementations.	Can change graph normalization, regularization, and optimization.
Feature source: Notebook A	DE features smoothed by LDS.	Consolidated de_lds MAT matrix.	Strongest input alignment with the paper.
Feature source: Notebook B	Benchmark described with DE+LDS.	Accepted file manifest points to PSD paths.	Changes the statistical feature representation.

Difference	Published / official description	Notebook implementation	Possible effect
Het architecture	Heterogeneous graph transformation and recurrent modeling.	Two normalized Graph-BiGRU streams with fusion.	Different capacity and inductive bias.
Het adjacency frequency	Implementation described as sample-specific.	One graph per 30-second clip.	All 30 samples share clip-level connectivity.
Adjacency sparsity	Not reported at this threshold.	4.47% global non-zero density; some clips retain only self-loops.	May reduce inter-channel message passing.
Hyperparameter behavior	Paper reports HetEmotionNet sensitivity at learning rate 0.01.	Notebook B selects 0.01 in nine folds.	Confirms architecture/optimization mismatch.
Hardware/software	RTX 3090 used for toolkit experiments.	Tesla T4; Python 3.12.13; PyTorch 2.10.0+cu128.	Runtime and numerical environment differ.
Original seed/checkpoints	Not released in the paper.	Seed 42 and notebook-specific checkpoints.	Exact trained-solution replication is impossible.

13. Runtime, Checkpointing, and Reproducibility Controls

Control item	Main notebook	HetEmotionNet-style notebook
Execution device	Tesla T4; CUDA verified	Tesla T4; CUDA verified
Software	Python 3.12.13; PyTorch 2.10.0+cu128	Python 3.12.13; PyTorch 2.10.0+cu128
Nested-search workload	3 models × 10 outer × 9 hyperparameter pairs × 3 inner folds	10 outer × 9 hyperparameter pairs × 3 inner folds
Final outer models	30 completed	10 completed
Checkpoint frequency	Every epoch with latest/previous fallback	Atomic last.pt every epoch and rolling ZIP bundle
Resume evidence	769 completed markers restored; 28 active checkpoints at resume stage	274 completed runs and 9 active checkpoints at resume start
Mean selected epoch	DGCNN 12.3; RGNN 75.7; SparseDGCNN 15.2	13.4
Exact total wall-clock runtime	Not stored in final notebook output	Not stored in final notebook output
Final exports	Outer-fold CSV, results, checkpoints, resume ZIP	Ten final checkpoints, pooled predictions, checkpoint ZIP, resume ZIP

The checkpoint designs are a strong operational contribution. They prevent completed inner folds from being retrained, preserve final outer models, and allow CPU-safe restoration. The principal missing reproducibility record is a final wall-clock breakdown for training, validation, and testing.

Conclusion

This report integrates completed ten-fold results from both uploaded notebooks. The HetEmotionNet-style model achieved the best mean cross-9 accuracy at 45.95 ± 3.53% and the best macro-F1 at 45.74 ± 3.40%. DGCNN followed with 40.43 ± 3.61% accuracy, SparseDGCNN with 39.36 ± 3.31%, and RGNN with 30.11 ± 2.70%. All four models performed substantially above the nine-class chance level of 11.11%.

The class-level evidence shows that DGCNN and SparseDGCNN share similar strengths and weaknesses, while RGNN exhibits pronounced class collapse despite much longer selected training. The HetEmotionNet-style model is more balanced and recognizes Amusement, Joy, and Fear most reliably, but its input source, graph construction, and architecture differ materially from the official method.

The final defensible statement is: the complete GNN4EEG cross-9 evaluation workflow was successfully reconstructed and executed for four graph-based models, with verified outputs from both notebooks. However, strict paper-level replication remains incomplete because exact official code identity, subject assignments, random seed, checkpoints, and the official HetEmotionNet processing pipeline are unavailable or were not reproduced exactly.

References

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[7] Group 09, "FACED MSGM Reproduction Report," uploaded report template.

[8] Uploaded implementation source: resumed-cross9.ipynb (completed DGCNN, RGNN, and SparseDGCNN exact-NCV outputs).

[9] Uploaded implementation source: resumed-cross-hetemotionalnet_Final.ipynb (completed HetEmotionNet-style outputs).